We observe that if two vectors are like vectors or unlike vectors, then the undirected lines (support) of the vectors are parallel to each other. There are pair of vectors which are neither like nor unlike vectors.

8.4 Algebra of Vectors

We have studied basic algebraic operations on real numbers and on matrices. Similarly we studied some operations on vectors. Now let us see how to add two vectors, subtract a vector from another vector and multiply a vector by a scalar.

8.4.1 Addition of Vectors

Let us define the sum of two vectors in two ways and see that they are the same. Let us assume that an object of unit mass is placed at the origin (0,0) in $\mathbb{R}^2$. We assume that the size of the object is just a point. Let us assume that two forces $\vec{a}$ and $\vec{b}$ of unit magnitude act on the object in the positive directions of x-axis and y-axis respectively (Fig.8.4). It is easy to guess that the object will move in the direction 45° to the x-axis as indicated in Fig.8.5. The forces $\vec{a}$ and $\vec{b}$ are equal to the vectors $\vec{a}$ and $\vec{b}$ as indicated in Fig. 8.6. We may think that the forces push the object in Fig. 8.4 and pull the object in Fig.8.6.

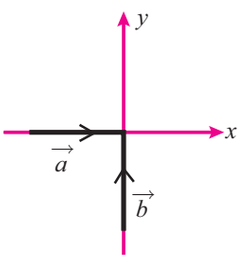


Fig. 8.4

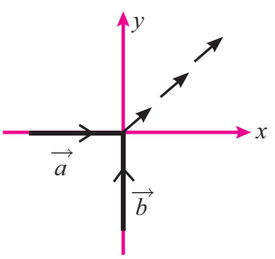


Fig. 8.5

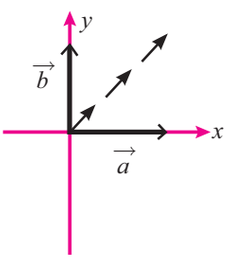


Fig. 8.6

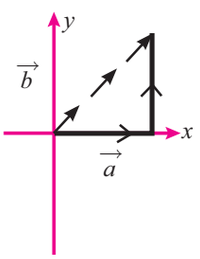


Fig. 8.7

The next question before us is 'How long will it go?'. Let us assume that the forces act one after the other. The force $\vec{a}$ will move the object one unit along the x-axis. So the object will move from (0, 0) to (1, 0). Now the force $\vec{b}$ will move the object vertically from (1, 0) to (1, 1). So finally the object will be at (1, 1) (Fig. 8.7). Thus the sum of the two vectors may be defined as the line segment joining (0, 0) and (1, 1) in the direction '(0, 0) to (1, 1)'.

Now, as in the same situation discussed above, let us assume that the force $\vec{a}$ has magnitude 2 instead of 1 (Fig. 8.8). It will not be difficult to guess that the object will move in a direction much closure to the x-axis as indicated in Fig. 8.9. Also we may guess that the object will go to the point (2,1). Thus the sum of the two vectors may be defined as the line segment joining (0,0) and (2,1) in the direction "(0,0) to (2,1)".

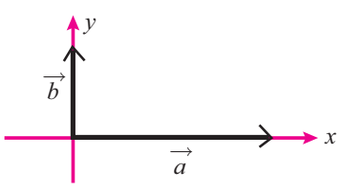


Fig. 8.8

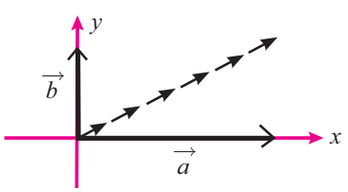


Fig. 8.9

In the two situations discussed above the directions of the forces are perpendicular to each other. This need not be the case in general. Even then we can add the forces by considering one after the other. For example let $\vec{a}$ and $\vec{b}$ be two forces in a plane as shown in Fig. 8.10.

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